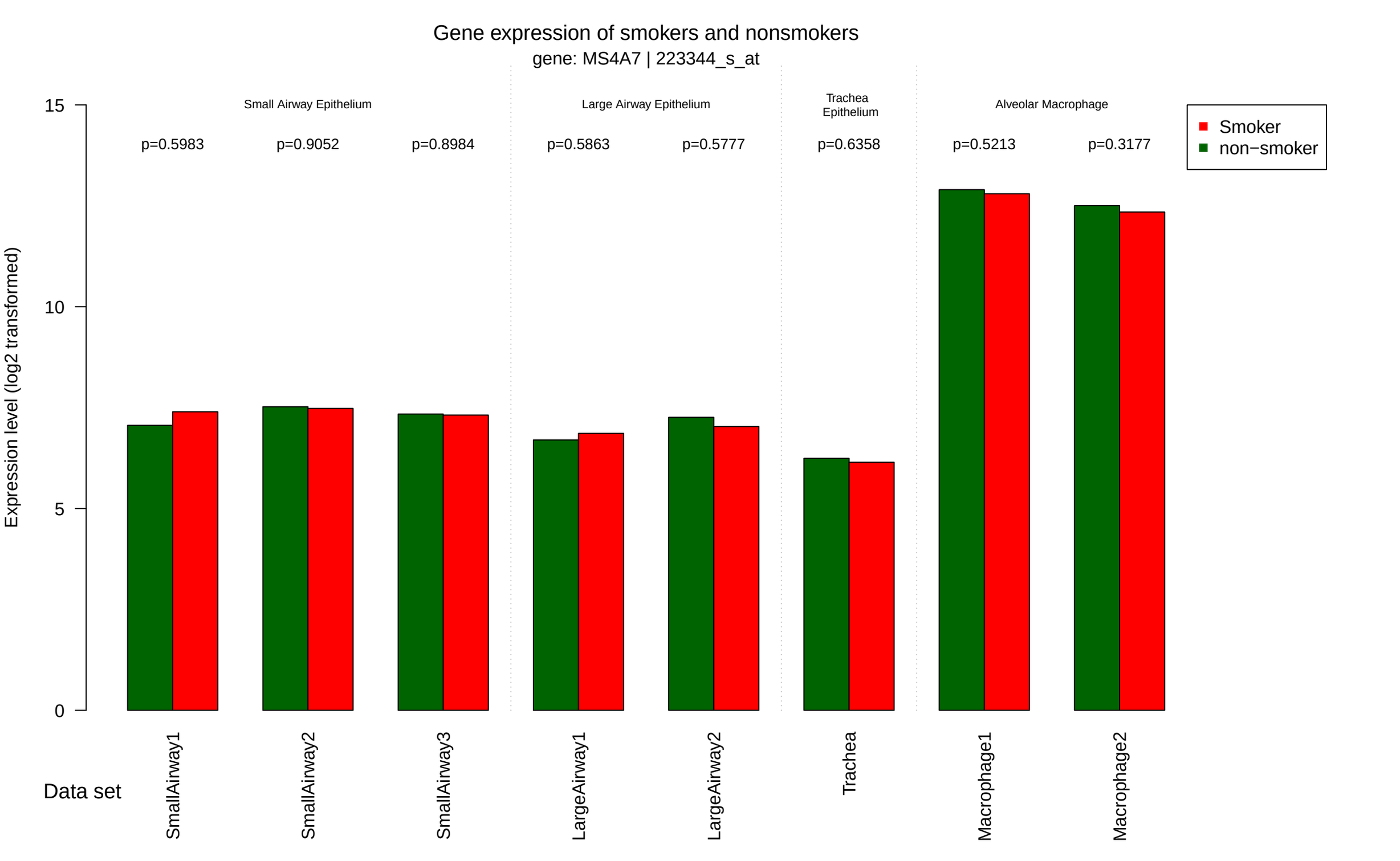
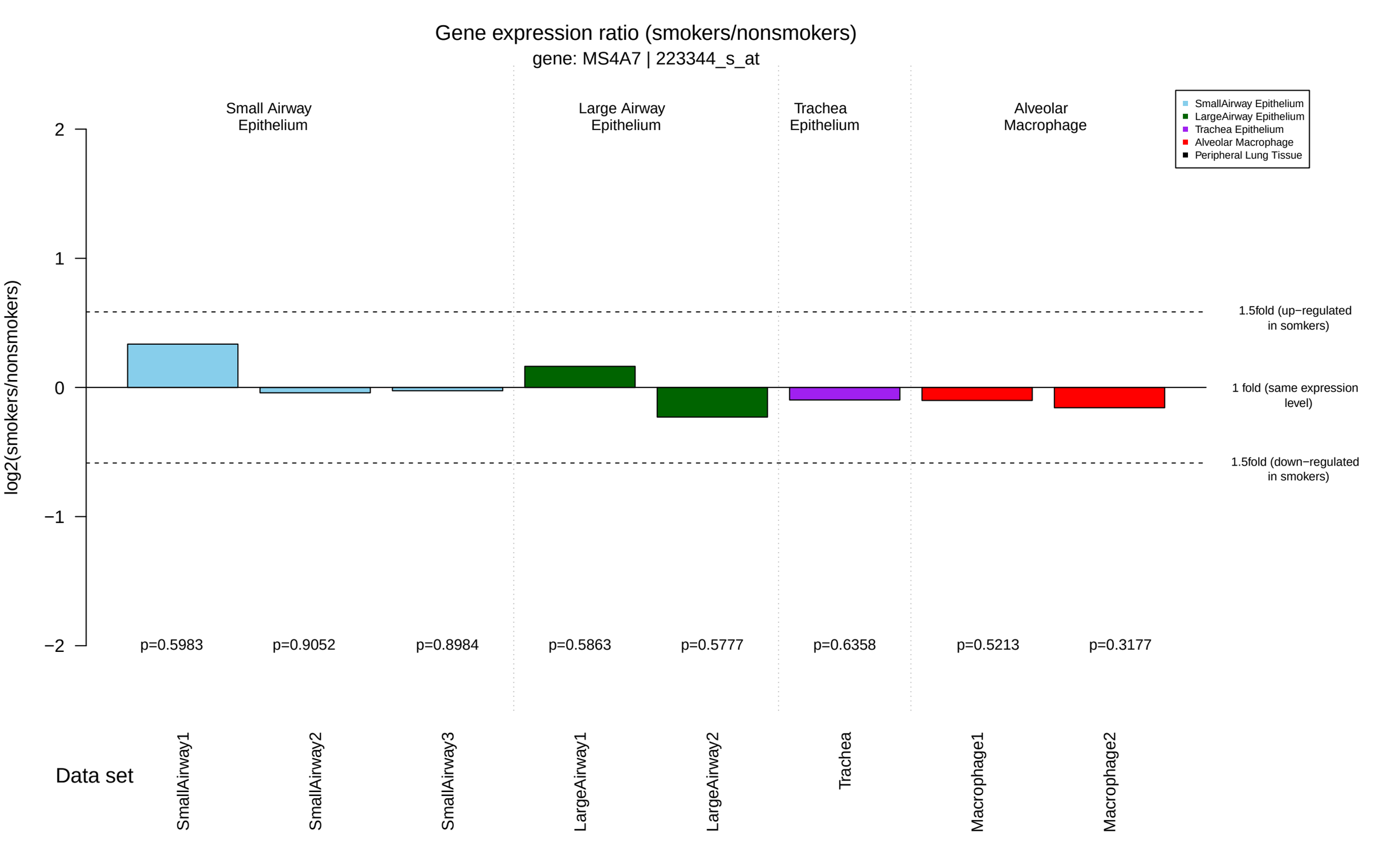
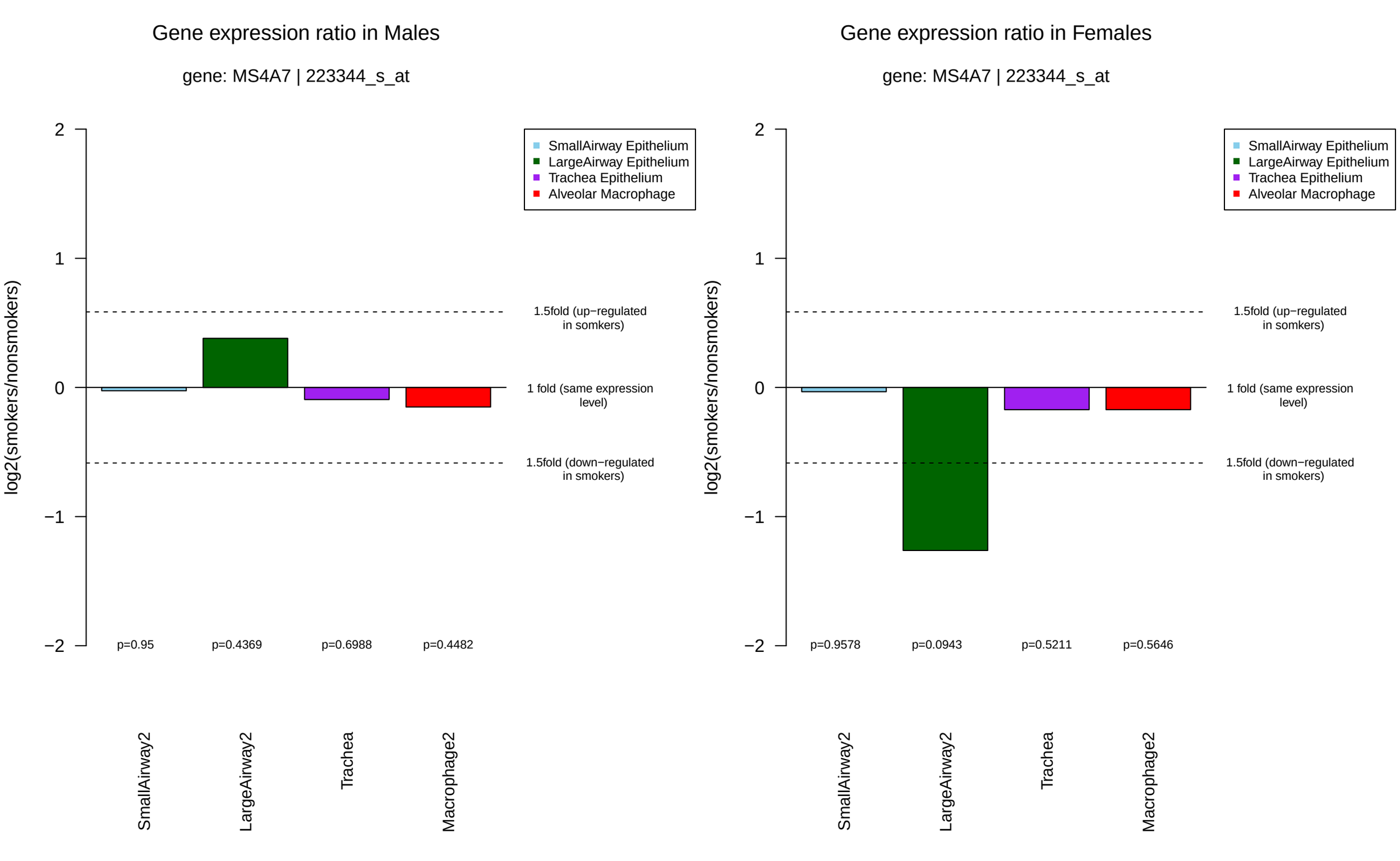
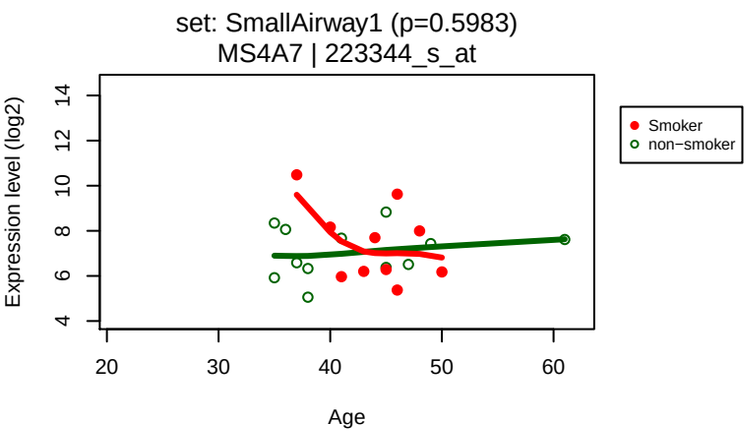
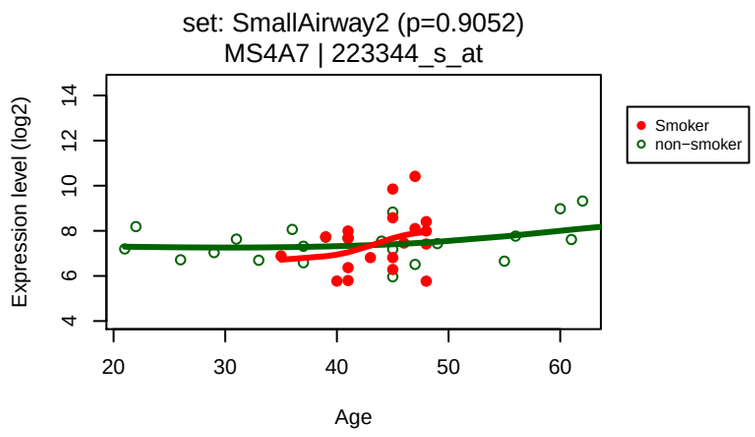
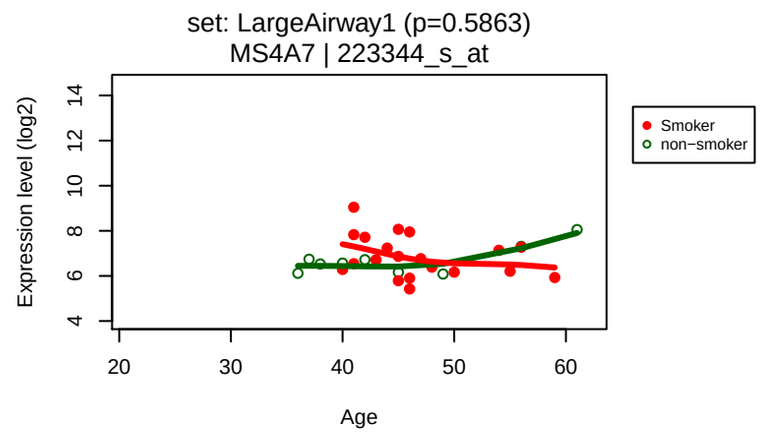

Gene expression of smokers and nonsmokers
gene: MS4A7 | 223344_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5983
p=0.9052
p=0.8984
p=0.5863
p=0.5777
p=0.6358
p=0.5213
p=0.3177
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

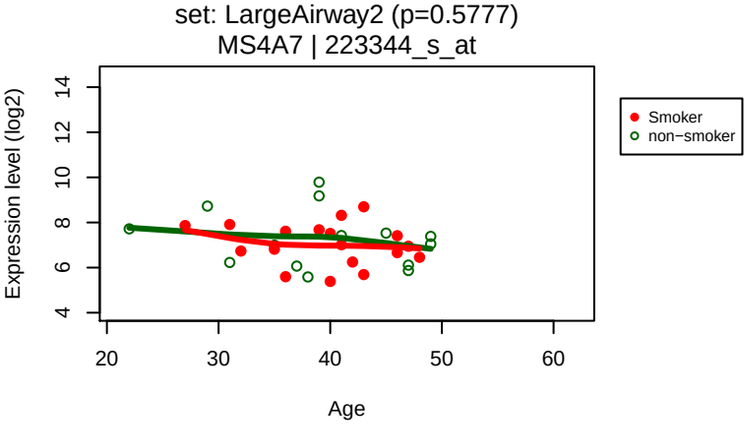

Gene expression ratio (smokers/nonsmokers)
gene: MS4A7 | 223344_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5983
p=0.9052
p=0.8984
p=0.5863
p=0.5777
p=0.6358
p=0.5213
p=0.3177
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

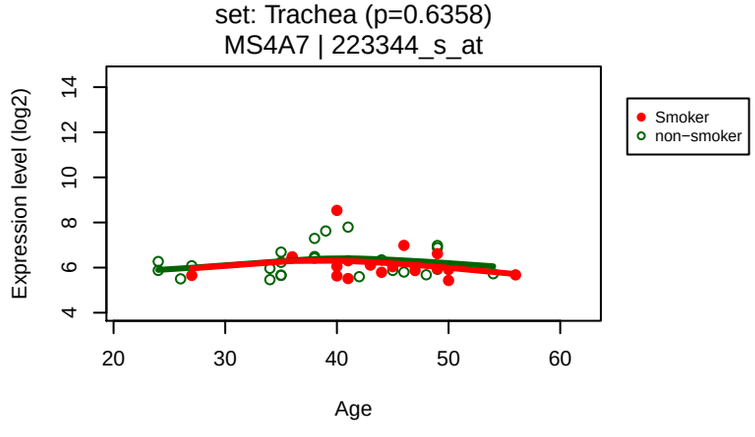

Gene expression ratio in Males
gene: MS4A7 | 223344_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.95
p=0.4369
p=0.6988
p=0.4482
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MS4A7 | 223344_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9578
p=0.0943
p=0.5211
p=0.5646
SmallAirway2
LargeAirway2
Trachea
Macrophage2

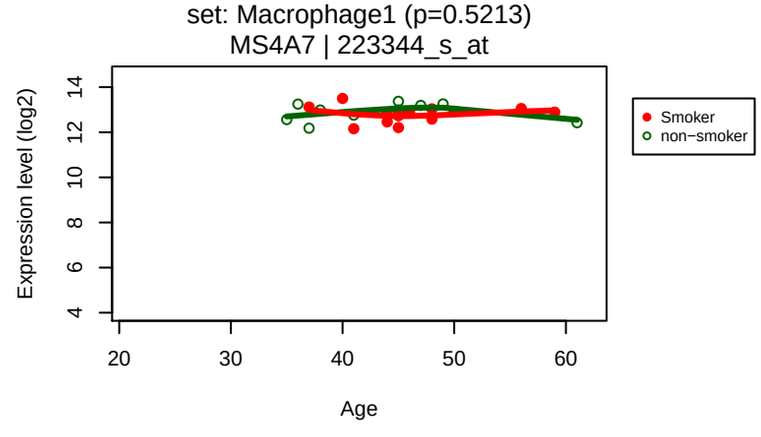

set: SmallAirway1 (p=0.5983)
MS4A7 | 223344_s_at
Expression level (log2)
Age
Smoker
non−smoker

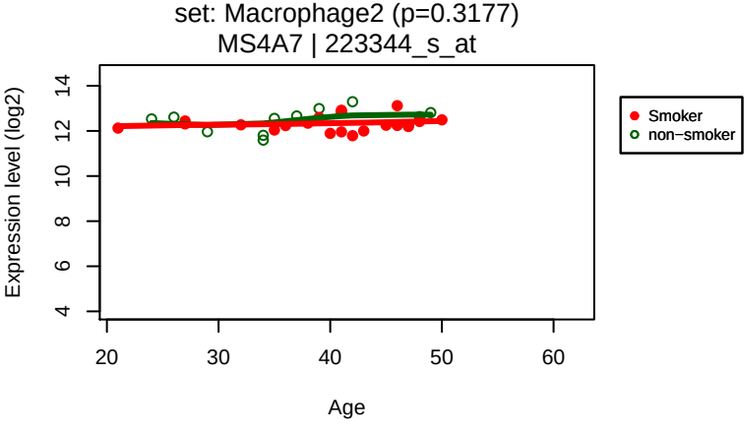

set: SmallAirway2 (p=0.9052)
MS4A7 | 223344_s_at
Expression level (log2)
Age
Smoker
non−smoker

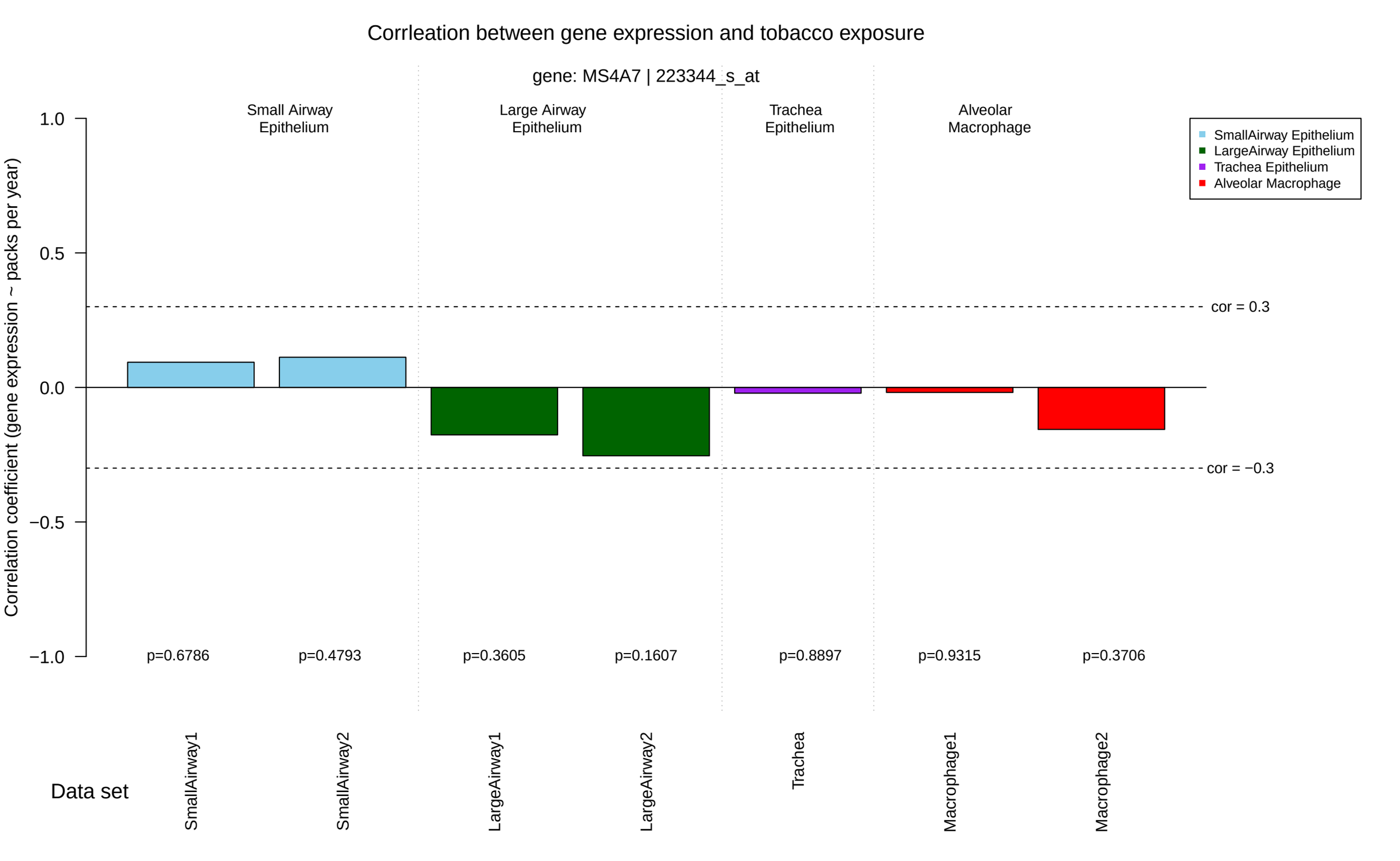

set: LargeAirway1 (p=0.5863)
MS4A7 | 223344_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5777)
MS4A7 | 223344_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6358)
MS4A7 | 223344_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5213)
MS4A7 | 223344_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3177)
MS4A7 | 223344_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MS4A7 | 223344_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6786
p=0.4793
p=0.3605
p=0.1607
p=0.8897
p=0.9315
p=0.3706
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2